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When Delivery Is **Late**, What Happens to **Customer** **Satisfaction?**

An **exploratory data analysis** project uncovering patterns in the e-commerce sector

Tools used

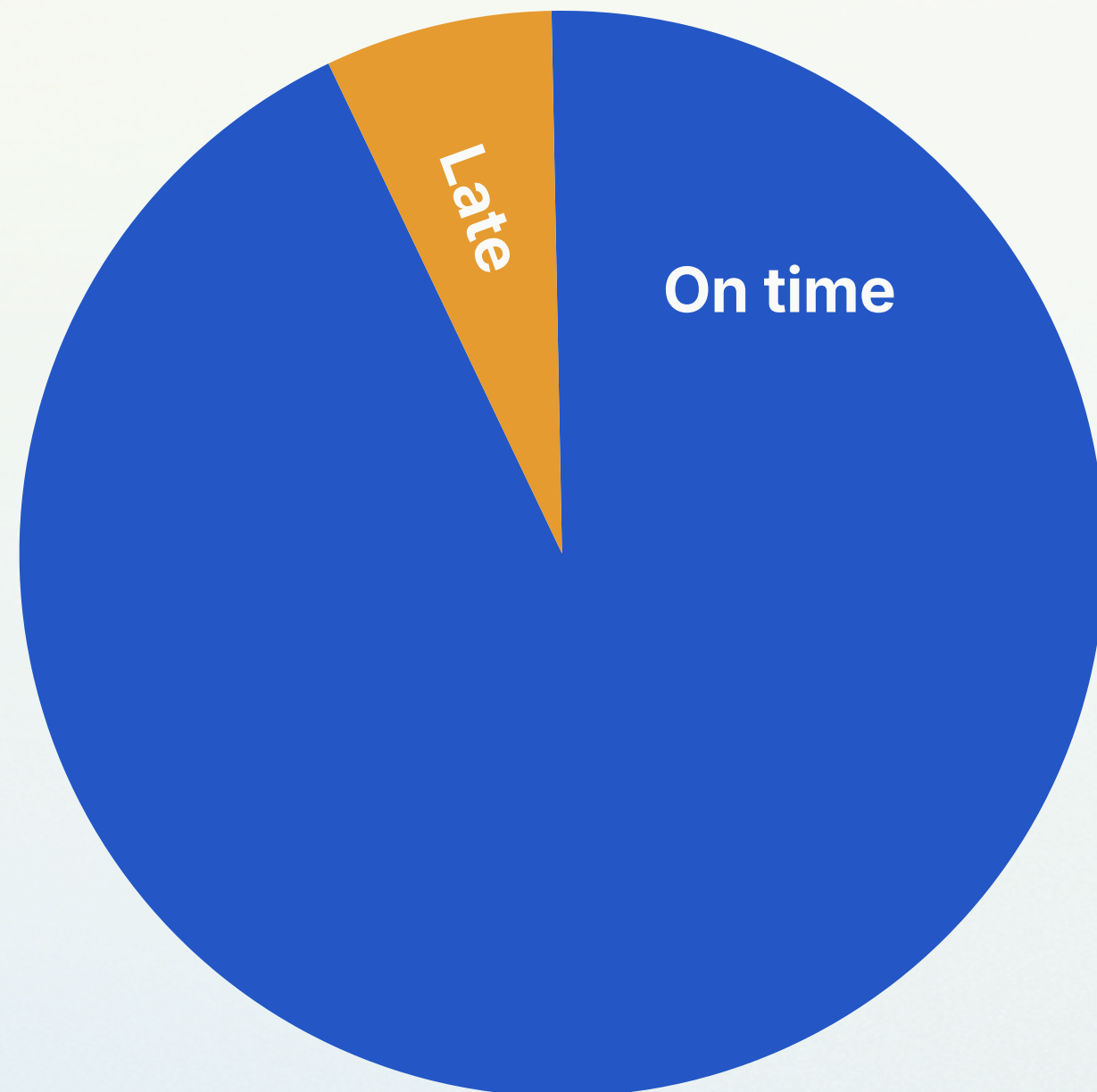


Brazilian E-Commerce Public Dataset by Olist · 2016–2018



Most Deliveries Arrive On Time. The Ones That Don't Tell a Different Story.

How often do late orders happen?



Out of
96,476

delivered orders, 

6.77%

were delivered **late** 

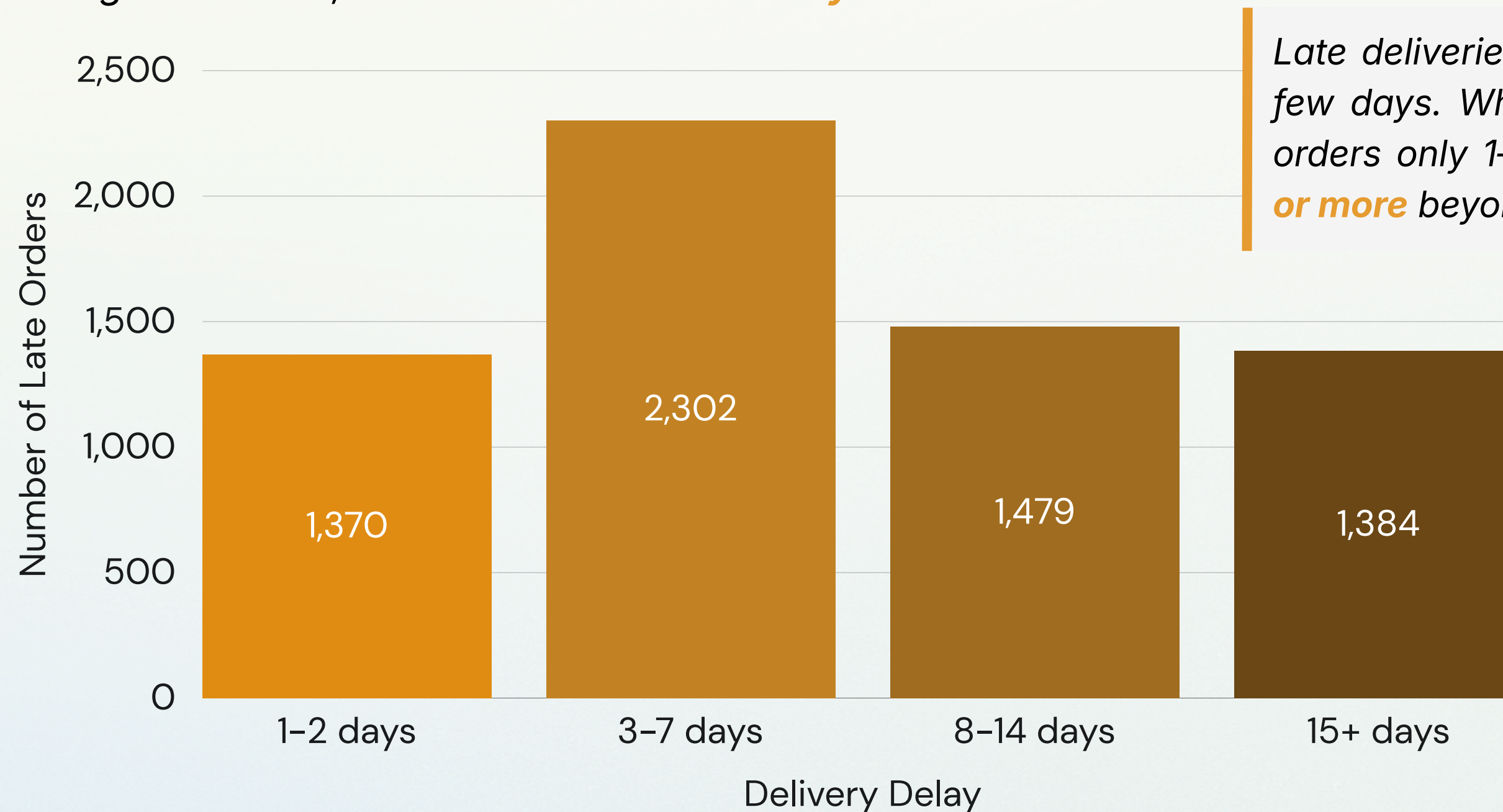
Only **1 in 15 orders arrived late**. But for those that did, the average delay was **more than 10 days**.

Among late orders, the average delay was **10.62 days**, with the longest observed delay reaching **188 days**.



Delays Are Usually Short. But a Meaningful Share Lasts for Weeks.

Among late orders, how severe were the **delays**?

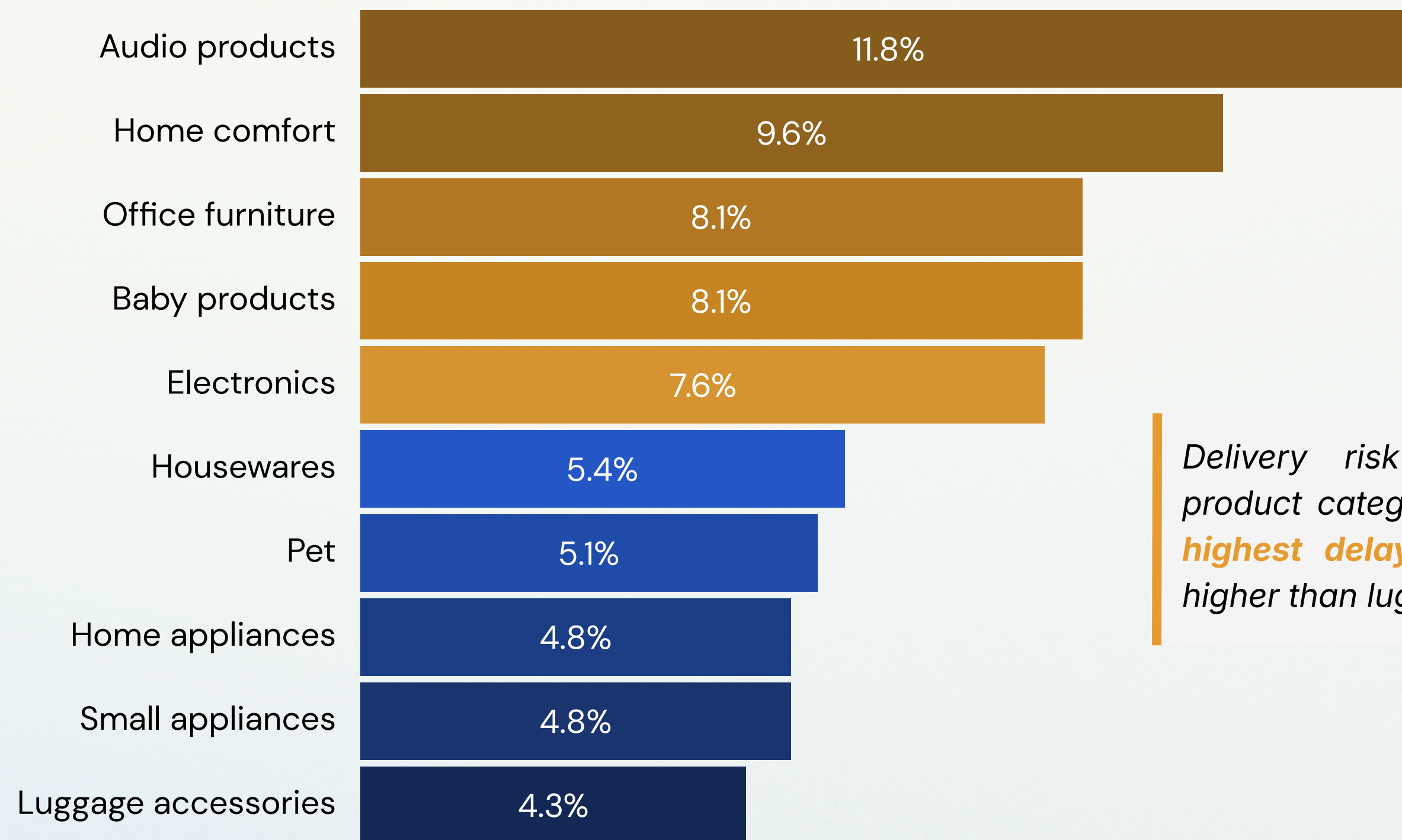


Late deliveries can last much longer than just a few days. While some customers received their orders only 1–2 days late, many **waited a week or more** beyond the promised date.



Some Product Categories Are **More Prone to Delays.**

5 highest delay rates vs 5 lowest delay rates among categories with ≥ 300 orders



Delivery risk **varies meaningfully** by product category. Audio products had the **highest delay rate** at 11.8%, nearly 3× higher than luggage accessories at 4.3%.



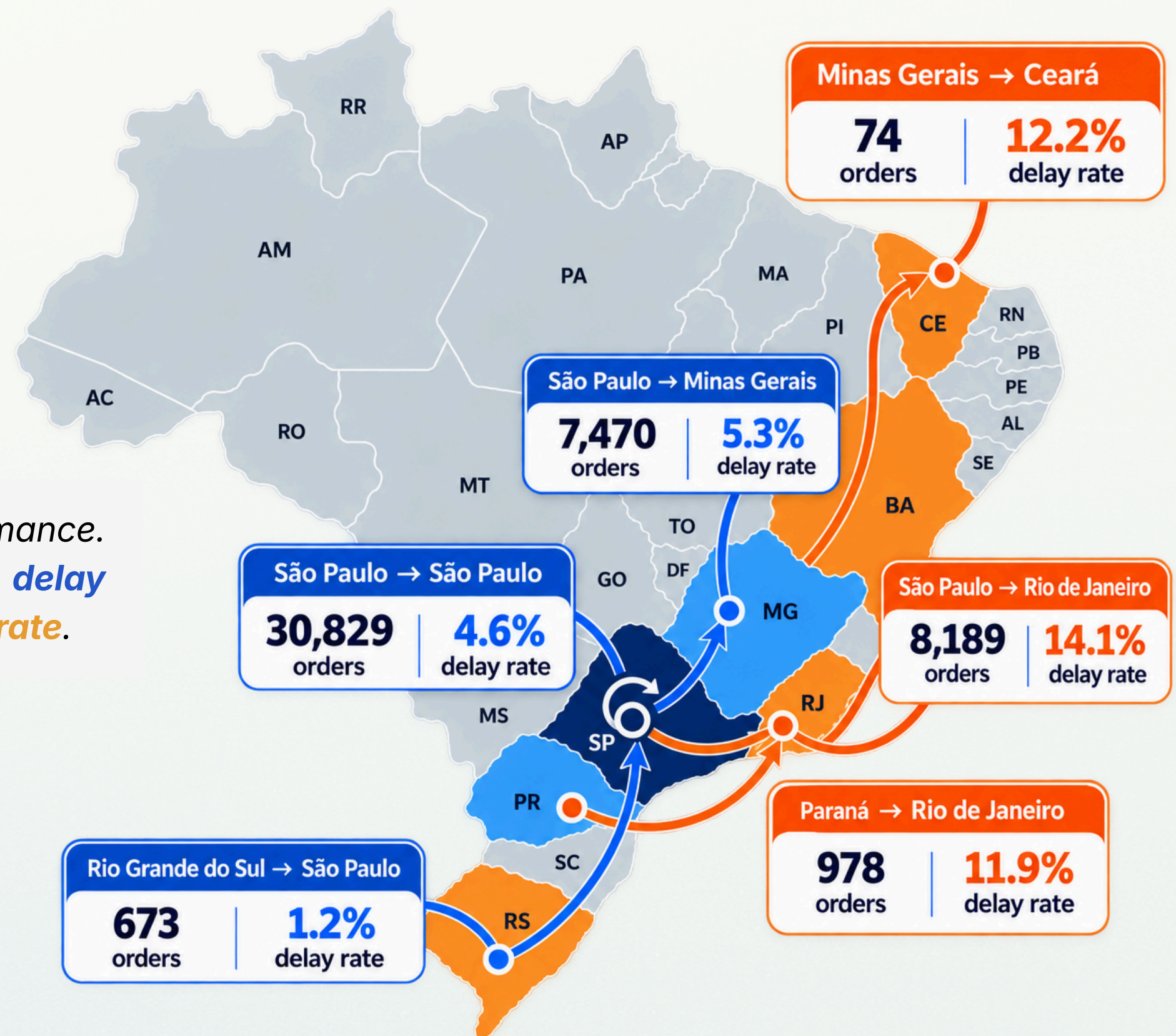
The Problem Gets Clearer When We Look at Routes.

Route volume alone doesn't explain delivery performance. Some of Olist's busiest corridors show **relatively low delay rates**, while others experience delays at **nearly 3× the rate**.

Other shipment characteristics show weaker patterns...

 Freight value	 Product weight	 Items per order
<R\$20 → 6.15%	<1 kg → 6.47%	1 item → 6.94%
R\$100+ → 9.90%	10+ kg → 8.84%	3 items → 4.90%

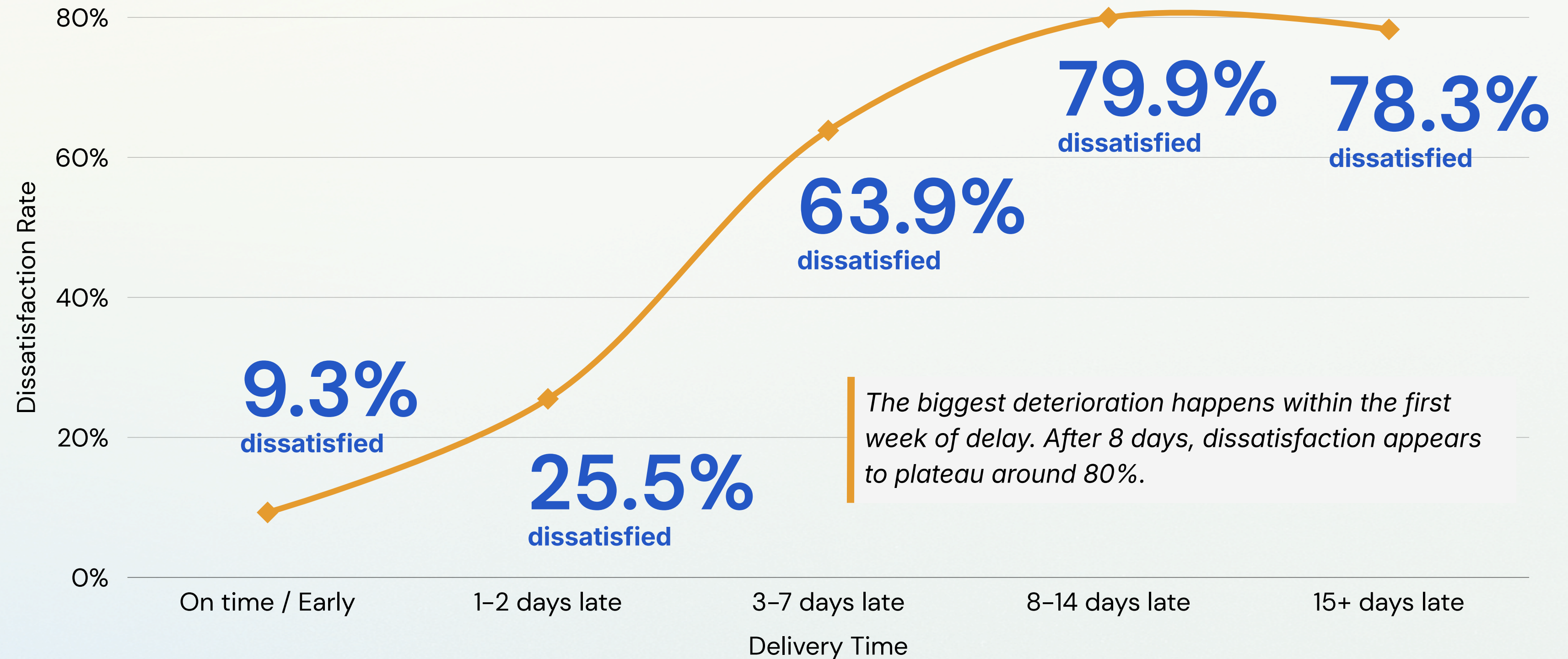
Order size by item count shows **no clear increasing pattern**.





Customer Dissatisfaction **Rises Sharply** After Just a Few Days of Delay.

Share of dissatisfied customers by delivery delay

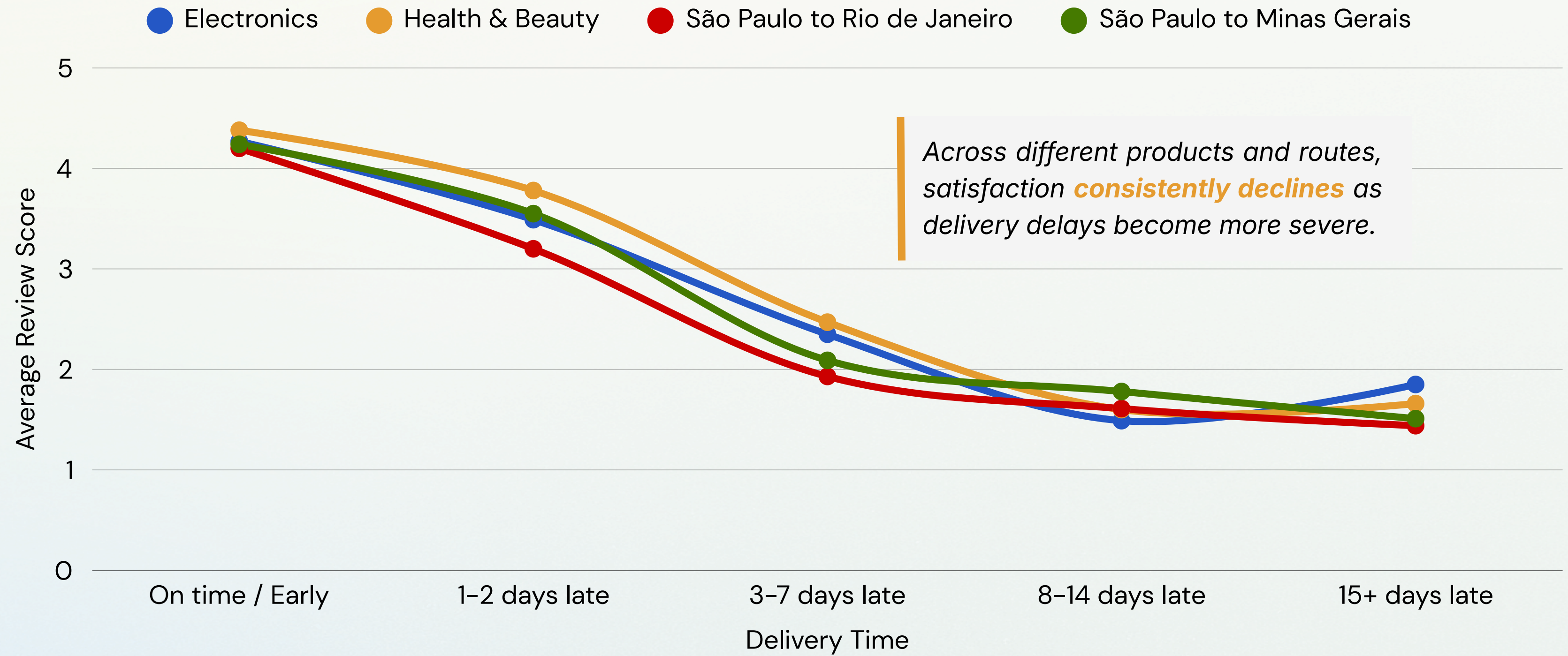


*Dissatisfaction = review score $\leq 2/5$. Results show association, not causation.



The Relationship Holds Across Categories and Delivery Routes.

Average review score based on product category and delivery routes





The Data Points to Targeted Delivery Improvements

A solid choice of action is to intervene where **delivery risk and customer impact overlap**.



Prioritize High-Risk Routes

Focus operational attention on high-volume corridors with disproportionately high delay rates.



Investigate Category-Specific Bottlenecks

Review fulfillment and logistics processes for categories with consistently higher delay rates, particularly audio, home comfort, office furniture, and baby products.



Protect the Customer Experience Early

Because dissatisfaction rises sharply within the first few days of delay, proactive communication or recovery measures should start before delays become severe.



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Thank You

Questions, feedback, or just curious about the analysis?



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